Soumodeep Karmakar

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Education

B.E. Electrical Engineering — 2024 – 2028 (Expected)
Jadavpur University, Kolkata — *Kolkata, India*

Skills

Languages & Core CS: Python, C/C++, JavaScript, TypeScript, SQL, OOP, Data Structures & Algorithms
AI & ML: Deep Learning, LLMs, Transformers, Mixture-of-Experts, NLP, Computer Vision, Multimodal Learning, Image Segmentation, Few-Shot Learning, RAG, Agentic AI, Context Engineering
Frameworks & Tools: PyTorch, Hugging Face, FastAPI, React, Pydantic AI, PostgreSQL, SQLite, asyncio, Docker, Modal, OpenCV, NumPy, Git

Experience

Research Contributor — Jun 2026 – Present
CMATER Lab, Jadavpur University (under Prof. Ram Sarkar) — *Kolkata, India*

- Built a **cross-dataset, few-shot medical image segmentation** pipeline using **SAM, CLIP, and CoOp** to generalize across unseen medical imaging domains.
- Developed **CLIP-guided mask selection** and pseudo-labeling for domain shift, contributing to **two papers** (one accepted at *CVIP 2026*, one first-author, under review).

Projects

Durable Tool-Execution Runtime | *Python, asyncio, SQLite* |

- Built a crash-recoverable execution backend for AI agent tool calls using a single-writer **SQLite WAL** thread, persistent commit-before-acknowledge guarantees, bounded backpressure, and argument-hashed idempotency.
- Implemented **epoch-fenced worker leases** with heartbeat renewal and policy-aware crash recovery, bridged to a ReAct agent for side-effect-free replay.
- Verified with **560 automated tests**, real process-kill (**SIGKILL**) crash-recovery chaos testing with zero lost jobs, and **2,115 jobs/s** sustained throughput (**3.63 ms** p99 admission latency).

ReAct Agent Framework | *Python, FastAPI, Modal, SWE-bench* |

- Built a modular **ReAct agent framework** with a code-debugging agent (sandboxed shell execution, git patch export) and a chess agent that runs **programmatic lookahead search** inside an isolated container.
- Designed a **working-memory context compaction** engine; resolved SWE-bench Django task django-15368 while cutting peak prompt size by **86.2%** (35.7K $\rightarrow$ 4.9K tokens) and total tokens by **86.3%**.
- Added an offline test suite, transient retry handling, and container sandboxing to isolate model-written code.

AI Customer Support | *FastAPI, React, TypeScript, PostgreSQL, Pydantic AI, Docker* |

- Built an automated Gmail support agent with **FastAPI**, **PostgreSQL**, **Pydantic AI**, and Docker for inbound triage, policy-grounded drafting, connection pooling, stale-job recovery, and duplicate-reply protection.
- Paired deterministic escalation rules and PostgreSQL full-text search with a **React + TypeScript** dashboard for ticket resolution and KB editing, backed by **39 offline tests** and an **18-case golden eval suite**.

nanoMoE | 120M params (65.8M active) | *PyTorch, MoE, MLA* |

- Built a **sparse Mixture-of-Experts LLM from scratch** with **DeepSeek-style Multi-head Latent Attention (MLA)** for KV-cache compression, RoPE, and SwiGLU.
- Designed a **Top-2 noisy-gating router** with auxiliary load-balancing loss (four experts per layer); pre-trained on **FineWeb-Edu** using Modal on H100 GPUs.

Publications

- **Calibrated SAM-CLIP Pseudo-Labeling and Cross-Dataset CutMix for Few-Shot Medical Image Segmentation.** **S. Karmakar**, S. Pani, S. Ghosh, J. Banerjee, R. Pal, R. Sarkar. *Knowledge-Based Systems (Elsevier).* **Submitted (under review)**.
- **Beyond Spatial Prototypes: Joint Spatial–Spectral Prototype Learning for Few-Shot Medical Image Segmentation.** J. Banerjee, R. Pal, S. Ghosh, S. Pani, **S. Karmakar**, R. Sarkar. *CVIP 2026.* **Accepted**.

Achievements

AISEHack 2026 – Top 4 Team, IIIT Hyderabad (IBM Theme: Flood Detection) — Apr 2026
Medha-Manthan 2025 – 1st Prize, IEEE Kolkata Chapter (assistive communication board) — Sep 2025
Hackforge – 3rd Position, Srijan'25, CodeClub Jadavpur University — Mar 2025